Sound Class 01 Flashcards

Generated by TrickFunda

1. What is the fundamental physical nature of sound?

’yM“Y(“ò ” .”)2 ’ÙL’I?‘R *”Ù0‘YC’I? ‘YM’ù> “™H?

Meaning:
Sound is a form of energy.

’yM“Y(“ò “ M‘É> ‘Y> •ù “ B’¢ 9”‰d

Trick / KD Hack
Logic Link: Energy cannot be created or destroyed, only transferred. When a physical force causes an object to vibrate, kinetic energy transfers into sound energy.
Logic Link: •©0”Ù “â ”² (’IK ’É(“é/“â “â 8‘Y$“â 9”, “ (“™@ ’‰7”Ù ‘Y?’ù> ‘É> “‰ ’I> “™H, ‘YG“Y2 “‰M’Y>’‰>•)

2. In what form is sound energy carried from one place to another?

’yM“Y(“ò “ M‘É> ‘YK •ù “‰M’Y>’, 8”r &”)8“ G “‰M’Y>’, $‘R “ù8 “ B’¢ .”y “)G ‘É>’ù> ‘É>’I> “™H?

Meaning:
It is carried in the form of waves.

•y8”r $“ ‘yK•" ”r 0”)* ’éG•" 2”r “é/“â “é$“â 9”‰d

Trick / KD Hack
Logic Link: A wave is a disturbance that transfers energy through a medium without permanently moving the matter itself. Sound relies on this moving disturbance.
Logic Link: ’I0•) •ù “Y?‘YM“yK’Ò 9”, ”² *’i>“ M’R ”² 8”Ù%“é/” 0”)* “‰G “‰M’Y>’‰>•)$“ ?’B “ù ’É?’‰> ’é>’yM’ù

3. What does the term 'Vibration' mean in physics?

’ÙL’I?‘Y@ ’éG•" y •)*’,r …f–' ation) “i,”Ù& ‘Y> ‘YM’ù> •Y0”Ù% “™H?

Meaning:
Vibration means Oscillation.

‘Y ’©(‘Y> •Y0”Ù% “™H ’iK“)((Oscillation)–@

Trick / KD Hack
Logic Link: Oscillation refers to a periodic back-and-forth motion around a central equilibrium point. Vibration is simply rapid oscillation.
Logic Link: ’iK“)(•ù ‘YG•)&”Ù0” / “‰ ’IA“)(’É?•)&”   ”r “é0”¹ ‘90 •ù •i5’y?‘R ‘yG-’©@‘¹G ‘Y@ ‘y$“ò ”² 8•)&“ M’Ù?

4. What are the two necessary conditions for the production and propagation of sound?

’yM“Y(“ò ”r ’IM’©>’i(‘I0 ’©M“ 8“é0 ‘YG “)?•ò &”² “Y6”Ù/‘R 6“ M’IG•" ”Ù/“â 9”‰ ?

Meaning:
1. Vibration of an object, 2. Presence of a medium.

1. ‘Y?“‰@ “Y8”Ù$” “â •)*’,Â "à ’é>’yM’ù. ‘Y@ •™*“‰M’Y?’I?–@

Trick / KD Hack
Logic Link: Sound transfers kinetic energy. Without an initial vibration, there is no kinetic energy. Without a medium (particles), there is nothing to bump into to carry that energy forward.
Logic Link: ’yM“Y(“ò ’I?‘Â “ M‘É> ‘YK “‰M’Y>’‰>•)$“ ?’B “ $” 9”‰d ’©M“ >“ ’Ù?‘R •)*’, ”r ,“ù(“âÂ ”¹ ‘y$“ù •©

5. If an explosion happens on the Moon, why does the sound never reach Earth?

’ù&“ò •)&”Ù0’é> ’©0 ‘YK•, 5“ù8”Ù+”¹ “™K’I> “™H, ’IK •™8‘Y@ ’yM“Y(“ò *”9%”Ù5” $‘R ’Ù@ ‘YM’ùK•" (

Meaning:
Because there is a vacuum between the Earth and the Moon, and sound cannot travel without a medium.

‘YM’ùK•) “ò *”9%”Ù5” “ •)&”Ù0’é> ‘YG ’É@‘¢ (“ù0”Ù5“é$ “™H, ‘I0 ’yM“Y(“ò ,“ù(“â “ù8” .“é’”Ù/’â ”r /“é

Trick / KD Hack
Logic Link: Space is a vacuum, meaning it lacks air or gas molecules. Since sound is a mechanical wave that requires particle collisions to move, it stops instantly where the atmosphere ends.
Logic Link: •Y ’I0“ù ”Ù7 •ù ’‰?“ M“Y>’B 9”,Â “ù8‘Y> •Y0”Ù% “™H ‘Y? •y8’éG•" 9“Y> ’ù> ‘yH“, ”r ’9A‘9 ‘Y> •Y-“é

6. If an explosion on the Moon makes no sound on Earth, why can we still see the flash of light from it?

’ù&“ò •)&”Ù0’é> ’©0 “™A•ò 5“ù8”Ù+”¹ ‘Y@ ’yM“Y(“ò *”9%”Ù5” *“ (“™@•" 8” (“é ’iG’I@, ’IK “™. ’¹?“ -”

Meaning:
Because light does not require any medium to travel; it can pass through a vacuum.

‘YM’ùK•) “ò *”Ù0‘Y>“b ”² /“é$”Ù0“â “ (”r ”r 2“ù ‘Y?“‰@ ’é>’yM’ù. ‘Y@ •i5“iM’ù ’I> ’‰9” “™K’I@ “™H;

Trick / KD Hack
Logic Link: Light is an electromagnetic wave. It propagates via oscillating electric and magnetic fields, not by bumping physical particles. Therefore, the vacuum of space is not a barrier.
Logic Link: ’©M“ “é6 •ù “Y?’iM’ùA’B ” .”Ù,‘Y@’ò $“ ‘r 9”‰d ’ù9 ’iK“)(“i@“" 5“ù&”Ù/” $ ‘I0 ‘©A•),‘Y@’ò ”Ù7”y$”Ù0

7. When watching a rocket launch from a distance, why do you see the rocket shining before you hear its roaring sound?

’iB“ 8”r 0”™ ”y “)I’‰M‘¢ &”y ’IG “‰.’òÂ ’© ”² 0”™ ”y ‘Y@ ‘y0”Ù ’‰> “‰A’‰(”r 8”r *“™2”r 0”™ ”y ‘©.‘Y

Meaning:
Because the speed of light is significantly greater (>>>) than the speed of sound.

‘YM’ùK•) “ò *”Ù0‘Y>“b ”   ’I? ’yM“Y(“ò ”   ’I? “‰G ‘Y>’¹@ •Y'“ù (>>>) “™K’I@ “™H–@

Trick / KD Hack
Logic Link: Both signals are emitted at the exact same time. But light travels at ~300,000 km/s while sound travels at just about 343 m/s in air. Light wins the race to your senses.
Logic Link: ’iK’‰K•" 8“ù ”Ù(“" ‘R 9” 8’é/ ’éG•" ’IM“‰0”Ù “ù$ “™K’IG “™H•)d “)G‘Y?’, *”Ù0‘Y>“b ã3 Ã   “ù.” ù8”y

8. What is the standard unit used to measure the loudness or noise of sound?

’yM“Y(“ò /“â 6”¹0 ‘Y@ ’©M“ ,“)$“â .“é*’‰G ‘YG “)?•ò *”Ù0’ùA‘YM’B .“é(‘R ‘Y>•, ”Ù/“â 9”ƒð

Meaning:
Decibel (dB)

’ G“‰?’É2 (Decibel - dB)

Trick / KD Hack
Logic Link: Loudness is subjective and relates to amplitude. To measure how our ears perceive this intensity exponentially, we use the logarithmic Decibel scale.
Logic Link: ’©M“ ,“)$“â 5”Ù/‘YM’I?’©0‘R 9”, “   ’ù>’â † × Æ—GVFR’ 8”r 8•),•)'“ù$ “™H–B /“' .“é*’‰G ‘YG “)?•ò “ò

9. Mathematically, what is a Decibel in relation to a 'bel'?

‘y#“ù$” / “ B’¢ 8”rÂ y,”y2' (bel) ‘YG “‰ ’É ’r .”y ’ G“‰?’É2 ‘YM’ù> “™H?

Meaning:
A Decibel is exactly 1/10th part of a bel.

’ G“‰?’É2 •ù ’ÉG“" “â ”   1/10“Y>•" 9“ù8”Ù8“â 9”‰d

Trick / KD Hack
Logic Link: In the metric system, 'deci' always means one-tenth. Just like a decimeter is 1/10 of a meter, a decibel is 1/10 of a bel.
Logic Link: ’é@‘ùM“ ?‘R *”Ù0’9>“)@ ’éG•"Â vFV6’r “â 9’éG“i> •Y0”Ù% ’i8“Y>•" 9“ù8”Ù8“â 9”¹$“â 9”‰d ‘É?“, $“ 9

10. What is the audible sound frequency range for normal human hearing?

“‰>’é>’‰M’ò .“é(“R 6”Ù0“Y# ‘YG “)?•ò 6”Ù0“YM’ò '”Ù5’‰? •i5”9$”Ù$“ò 8” .“â ”Ù/“â 9”ƒð

Meaning:
20 Hz to 20 kHz (20,000 Hz).

20 “™0”Ù ”Ù (Hz) “‰G 20 ‘Y?“)K“™0”Ù ”Ù “Â †´‡¢’ ƒ# Ã   ‡¢™d

Trick / KD Hack
Logic Link: Frequencies below 20 Hz (Infrasound) are too slow to trigger our ear structure, and those above 20 kHz (Ultrasound) vibrate too rapidly for our eardrums to decode.
Logic Link: 20 “™0”Ù ”Ù (•y(”Ù+”Ù0“é8“é •)!) “‰G ‘Y. •i5”9$”Ù$“ù/“é “™.“é0”r “é(‘Y@ “‰ “ ’‰> ‘YK ‘ùM“ ?‘y0

11. What is the 'Persistence of Hearing' time for the human ear?

’é>’‰5 ‘Y>’, ”r 2“ù "‘iM“ 5’2 ” &”9"“É$“âr … ersistence of Hearing) ‘Y> “‰.’ò ”Ù/“â 9”ƒð

Meaning:
1/10 seconds (or 0.1 sec).

1/10 “‰G‘Y ’ ‰/“â ã 8”y •)!)–@

Trick / KD Hack
Logic Link: The brain needs a tiny fraction of a second to process a sound. If two sounds hit the ear faster than 0.1s apart, the brain merges them into one continuous sound.
Logic Link: ’é8”Ù$“ù7”Ù ‘YK •ù ’yM“Y(“ò ”² 8•)8“é"“ù$ ‘Y0’‰G ‘YG “)?•ò ‘R 8”y •)! ‘YG •ù ‘¹K‘ùG “‰G •Y “b ” ‘

12. What is the 'Persistence of Vision' time for the human retina (eye)?

’é>’‰5 “ G‘ù?’‰> (•i ‘b’ ”r 2“ù "iC“yM‘ù? ‘Y@ ’iC’)<’I>’ (Persistence of Vision) ‘Y> “‰.’ò ”Ù/“â 9”ƒð

Meaning:
1/16 seconds (or 0.062 sec).

1/16 “‰G‘Y ’ ‰/“â ã c" 8”y •)!)–@

Trick / KD Hack
Logic Link: To see smooth motion, frames of light must be shown faster than the retina can discard the old image. If images change faster than 1/16th of a second, it creates the illusion of a continuous video.
Logic Link: “‰A‘©>“ B ‘y$“ò &”y ’‰G ‘YG “)?•òÂ *”Ù0‘Y>“b ”r +”Ù0”y. ‘YK “ G‘ù?’‰> ’iM“Y>“ > ’©A“ >’‰@ ‘¹5“ò

13. What is the scientific study of the production of sound waves called?

’yM“Y(“ò $“ ‘yK•” ”r ’IM’©>’i(‘YG “YH‘ÉM‘é>’‰?‘R ’yM’ù/’, ”² ”Ù/“â “™> ‘É>’I> “™H?

Meaning:
Acoustics

’yM“Y(“ù ” „ 6÷W7F–72•

Trick / KD Hack
Logic Link: Acoustics is the branch of physics covering mechanical waves in gases, liquids, and solids, specifically focusing on sound.
Logic Link: •ù ”™8”Ù “ù ”Ù8 ’ÙL’I?‘Y@ ‘Y@ “Y9 “i>‘i> “™H ‘ÉK ‘yH“‰K•"Â $“ 2 ’©&“é0”Ù%”¹ ‘I0 ’ K“, *’i>“ M’YK•"

14. Which instrument is used for the study and visual display of the waveform of any signal?

‘Y?“‰@ ’Ù@ “‰ ‘YG’B ‡6–væ Â’ ”r $“ ‘r 0”)* (waveform) ‘YG •Y"”Ù/’ù(‘I0 ’iC“iM’ò *”Ù0’i0”Ù6’, ”r 2“ù ‘

Meaning:
Oscilloscope

‘ 8“ù2”¹8”Ù ”¹* (Oscilloscope)

Trick / KD Hack

Logic Link: An oscilloscope physically plots the electrical representation of a wave on a graph over time, showing its true oscillation shape.
Logic Link: •ù •i8”Ù “‰@“)8”Ù ’¢ 8’é/ ‘YG “‰>’R ‘R ”Ù0“é+ ’©0 •ù ’I0•) ‘YG “Y?’iM’ùA’B *”Ù0’I?’‰?’y?’IM“R ”

15. Based on their requirement for a medium, what are the two main categories of waves in physics?

’é>’yM’ù. ‘Y@ •i5“iM’ù ’I> ‘YG •i"“é0 ’©0, ’ÙL’I?‘Y@ ’éG•" $“ ‘yK•" ” &”² .” ”Ù/ “iM“ G’9?’ù>•" ”Ù/“â 9”%

Meaning:
1. Mechanical Waves
2. Non-Mechanical (Electromagnetic) Waves

1. ’ù>•)$”Ù0“ù ’I0•) ”y (Mechanical Waves)
2. ‘yH“ Ù/“é ’IM“ ?‘R ‰5“ù&”Ù/” $ ‘©A’éM’É ” /) ’I0•) ”y (Non-Mechanical Waves)

Trick / KD Hack

Logic Link: Nature divides energy transmission into two regimes: one that uses actual matter as a vehicle (mechanical) and one that is self-propagating pure energy fields (electromagnetic).
Logic Link: ’©M“ ”9$“ò “ M‘É> “‰ ‘©0’2 ”² &”² 5”Ù/“Y8”Ù%“é •" .”y “Y?’Ù>‘É?’B “ $” 9”ƒ •ù ‘ÉK “Y>“‰M’I5“ù

16. What is the defining characteristic of Mechanical Waves?

’ù>•)$”Ù0“ù ’I0•) ”¹ (Mechanical Waves) ‘Y@ ’©0“ù-“é7“ù$ “Y?“iG“y$“â ”Ù/“â 9”ƒð

Meaning:
They explicitly require a material medium to travel.

•™(”Ù9”y ’ù>’IM“ > ‘Y0’‰G ‘YG “)?•ò 8”Ù*“yM‘ò 0”)* “‰G •ù ’ÙL’I?‘R .“é"’Ù/’â ” “Y6”Ù/‘Y$“â 9”¹$” 9”

Trick / KD Hack

Logic Link: 'Mechanical' means physical parts are working together. The 'parts' here are molecules bumping into one another. No molecules = no mechanics.
Logic Link: ''ù>•)$”Ù0“ù ' ‘Y> •Y0”Ù% “™H ’ÙL’I?‘R -“é •ù “‰>’R “é. ‘Y0 “ 9”r 9”‰ –B /“™>• y-“é ' •ù -’iB“‰0”r

17. What is the defining characteristic of Non-Mechanical Waves?

‘yH“ Ù/“é ’IM“ ?‘R $“ ‘yK•" „æöâÔÖV6† æ–6 Â Waves) ‘Y@ ’©0“ù-“é7“ù$ “Y?“iG“y$“â ”Ù/“â 9”ƒð

Meaning:
They do NOT require a material medium and can travel through a vacuum.

•™(”Ù9”y ‘Y?“‰@ ’ÙL’I?‘R .“é"’Ù/’â ” “Y6”Ù/‘Y$“â (“™@•" 9”¹$” 9”, “ 5”r (“ù0”Ù5“é$ “‰G “™K‘Y0 ’

Trick / KD Hack

Logic Link: Since they are composed of oscillating electric and magnetic fields, they sustain themselves without needing atoms to push against.
Logic Link: ‘©B•) “ò 5”r &”¹2’‰6” 2 “Y?’iM’ùA’B “ ” ’É ” / ‘YM“yG’IM“ K•" 8”r ,’‰G “™K’IG “™H•"Â 5”r *“ .“é#” •

18. List 6 common examples of Non-Mechanical (Electromagnetic) Waves.

‘yH“ Ù/“é ’IM“ ?‘R ‰5“ù&”Ù/” $ ‘©A’éM’É ” /) ’I0•) ”¹ ‘YG 6 “‰>’é>’‰M’ò ’i>“™0’2 &”y –@

Meaning:
1. Light
2. Microwave
3. Infrared
4. X-rays
5. UV Rays
6. Radio Waves

1. ’©M“ “é6 (Light)
2. ’é>•y ”Ù0”¹5”y5 (Microwave)
3. •y(”Ù+”Ù0“é0”y! (Infrared)
4. •ù ”Ù8-“ G (X-rays)
5. ’ùB“Y@ ‘Y?“ #”y (UV Rays)
6. “ G’ ?’ùK ’I0•) ”y (Radio Waves)

Trick / KD Hack
Logic Link: All these are simply different manifestations of EM energy at varying frequencies. Because they are the same core phenomenon, none of them need a medium.
Logic Link: ’ùG “‰-” “) -•Y2‘r “YC’IM’I?’ùK•" *“ •ù. •©0”Ù “â ” “) -•Y2‘r ’Ù?“YM’ù ”Ù$“ù/“é “™H•)d ‘YM’ùK•) “

19. What are the two sub-types into which Mechanical Waves are divided based on particle vibration direction?

‘Y# ‘Y ’©(’i?“i> ‘YG •i“‘é0 ’©0 ’ù>•)$”Ù0“ù ’I0•) ”¹ ‘YK ‘Y?’, &”² ’¢Ù*”Ù0‘Y>“ K•" .”y “Y?’Ù>‘É?’B “ù/“â

Meaning:
1. Longitudinal Waves
2. Transverse Waves

1. •Y(” &”‰0”Ù‘”Ù/ ’I0•) ”y (Longitudinal Waves)
2. •Y(” *”Ù0“‰M’R $“ ‘yG•" …@ransverse Waves)

Trick / KD Hack
Logic Link: Mechanical waves depend on atoms bumping each other. They can either bump forward/backward in the direction of travel (Longitudinal) or shake side-to-side/up-and-down (Transverse).
Logic Link: ’ù>•)$”Ù0“ù ’I0•) ”y ’©0’é>’9A‘9 ‘YG •ù -’iB“‰0”r 8”r ‘Y0“é(”r *“ (“ù0”Ù-“ “ $” 9”‰ –B 5”r /“â $”² /“é

20. In a Longitudinal Wave, what is the geometric relationship and angle between the direction of wave travel and the vibration of particles?

•ù •Y(” &”‰0”Ù‘”Ù/ ’I0•) (Longitudinal Wave) ’éG•"Â $“ ‘r /“é$”Ù0“â ” &“ù6“â “ ’9K•" ”r •)*’, ”r ,” ‘É

Meaning:
The particles vibrate Parallel (L & R) to the direction of the wave. The angle (theta) is 0°.

‘Y# ’I0•) ‘Y@ ’i?“i> ‘YG “‰.“é(“é ’I0 (L ‘I0 R) ‘Y ’©(‘Y0’IG “™H•)d ‘YK’2 ‰%” “â’ 9”‰d

Trick / KD Hack
Logic Link: 'Longitudinal' implies 'along the length'. If the wave pushes forward, the particles also squeeze forward and bounce backward exactly on that same line.
Logic Link: '•Y(" &"‰0"Ù'"Ù/' 'Y> 'I>'IM'©0"Ù/ '") 'É>•, "r 8"é%' "™H–B /'i? 'I0•) •i "r " " "YG")$" 9",Â $"² '2 -"

21. Give two key examples of Longitudinal Waves mentioned in the notes.

'‰K'ùM", ."y •™2"Ù2"ù "ù$ •Y(" &"‰0"Ù'"Ù/ 'I0•) "¹ (Longitudinal Waves) 'YG 'iK '©M" ." •™&"é9" # 'i

Meaning:
1. Sound Waves
2. Spring Waves

1. 'yM"Y("ò $" 'yG•" …6÷VæB Waves)
2. "‰M'©M" ?•) 'I0•) "y (Spring Waves)

Trick / KD Hack
Logic Link: Both sound and compressed springs push matter forward, which rebounds backward, operating completely parallel to the direction of travel (0 degrees).
Logic Link: 'yM"Y("ò " 8•)*" !"É?'B 8"Ù*"Ù0"ù 'r &"¹("¹ '©&"é0"Ù% 'YK •i "r "YG")$"r 9"‰ , 'ÉK '©@'¹G 'Y@ '90 •

22. In a Transverse Wave, what is the geometric relationship and angle between the direction of wave travel and the vibration of particles?

•ù •Y(" *"Ù0"‰M'R $" 'r …@ransverse Wave) 'éG•"Â $" 'r /"é$"Ù0"â " &"ù6"â " '9K•" "r •)*', "r ," '

Meaning:
The particles vibrate Perpendicular (Up & Down) to the direction of the wave. The angle is 90°.

'Y# 'I0•) 'Y@ 'i?"i> 'YG ") 'É5'B ‰ '©0 'I0 '‰@'©G) 'Y '©('Y0'IG "™H•)d 'YK'2 " 9"‰d

Trick / KD Hack
Logic Link: 'Trans' means 'across'. The energy travels forward horizontally, but the particles are simply displaced vertically (across the path) to pass the energy along.
Logic Link: "ùM" >•)8' 'Y> •Y0"Ù% "™H '•i0-'©>" yd •©0"Ù "â "Ù7"‰$"ù " B'¢ 8"r 'yG 'É""É$" 9",Â 2"y "ù(•©0"Ù

23. What is the primary example of a Mechanical Transverse Wave given in the notes?

'‰K'ùM", ."y 'i?•ò •ò /"é 'IM" ?'R '‰A'©M" 8"Ù% 'I0•) (Mechanical Transverse Wave) 'Y> '©M" >'Y."ù

Meaning:
Water Waves

'É2 'I0•) "y (Water waves)

Trick / KD Hack
Logic Link: If you place an object on a pond and a water wave passes, the object bobs Up and Down (90 degrees to wave direction) without moving forward.
Logic Link: 'ù&"ò '¢ $"é2"é, 'éG•" "¹ "Y8"Ù$" 0'i$"r 9"‰ 'I0 '©>'‰@ 'Y@ ")9" " " $" 9",Â $"² 5"‰M'IA •i "r ,')<"

24. Combining its properties, mathematically define the physical nature of a Light wave.

•y8‘YG ‘yA’9K•" ”2 .“ù2“é “ Â *”Ù0‘Y>“b „Æ–v‡B’ $“ ‘r ” -”É$“ù ’©M“ ”9$“ò ”2 ’9?’I@’ò 0”)* “‰G ’©0“ù

Meaning:
Light is both Transverse AND Non-Mechanical.

’©M“ “é6 •Y(” *”Ù0“‰M’R …@ransverse) ‘I0 ‘yH“ Ù/“é ’IM“ ?‘R „æöâÔÖV6† æ–6 Â’ &”1(”1 “™H–@

Trick / KD Hack
Logic Link: It doesn't need a medium (Non-Mechanical), but its electric/magnetic fields oscillate at 90° to the path of travel (Transverse).
Logic Link: •y8”r “ù8” .“é'”Ù/’â ” “Y6”Ù/‘Y$“â (“™@•" 9”1$” 9”, ‰ ”‰0-’ù>•)$”Ù0“ù), “)G‘Y?’, “‰ ”r 5“ù&”Ù/” $/

25. When a wave travels through a Spring, what are the two alternating phases created?

‘É, ‘YK•, $“ ‘r “ù8” 8”Ù*”Ù0“ù ‘r 8”r 9”1 “ ” “ $” 9”,Â $”2 ,’‰(”r 5“é2”r &”2 ‘Y>•)$“ “ # ‘YM’ù> “™H•#ð

Meaning:
Compression and Expansion (or Rarefaction).

“‰ ’©@’ <’, „6ö× &W76–öâ’ “ 5“ù8”Ù$“é0 / “Y?“ 2’, „W‡ ç6–öâ ò & &Vg action)–@

Trick / KD Hack
Logic Link: In longitudinal waves, energy transfers by squeezing particles tightly together (Compression) and then pulling them apart as the energy passes (Rarefaction).
Logic Link: •Y(” &”‰0”Ù'”Ù/ ’I0•) ”1 ’éG•"Â “ M‘É> ‘Y#”1 ‘YK •ù “‰>’R “‰ “ (“ù ”1!“É “ ‰8•)*” !“É() “‰M’Y>’‰>

26. In a spring undergoing a longitudinal wave, how does the direction of the wave relate to the direction of vibration mathematically?

•ù •Y(” &”‰0”Ù'”Ù/ ’I0•) “‰G ‘yA‘É0’‰G “Y>“)G “‰M’©M“ ?•) ’éG•"Â $“ ‘r ” &“ù6“â ’9?’I@’ò 0”)* “‰

Meaning:
They are in opposite directions but on the same line (0 degrees).

“YG “Y?’©0” $ ’i?“i>‘9 ’éG•" 9”‰ “)G‘Y?’, ‘R 9” 0”y “â ƒ !“ù ”Ù0” ’ *“ 9”‰ –@

Trick / KD Hack
Logic Link: The particle vibrates back and forth. When bouncing back (opposite to wave forward motion), it remains exactly on the 0° line. It never deviates off-axis.
Logic Link: ‘Y# •i ”rÙ*” ”r •)*’, “ $“â 9”‰d “Y>’©8 •™ “)$”r 8’é/ (’I0•) ‘YG •i ”r ” ’I? ‘YG “Y?’©0” $), ’ù9 ’ @‘R 0

27. What type of wave is produced when playing a Dholak (Indian drum)?

’)K“) ’É “é$”r 8’é/ ‘Y?“, *”Ù0‘Y>“ ” $“ ‘r ’IM’©(”Ù(“™K’I@ “™H?

Meaning:

Longitudinal Wave

•Y(” &”‰0”Ù'”Ù/ ’I0•) (Longitudinal Wave)

Trick / KD Hack
Logic Link: Striking the Dholak skin pushes it inward and outward. This flat pushing motion creates compressions and rarefactions in the adjacent air molecules, creating a sound wave (which is longitudinal).
Logic Link: ’)K“) ‘Y@ ‘i>“"' *“ *”Ù0“™>“   “ (”r 8”r 5“' •)&“   “ ,“é9“   ”   “ '‘YG“)$“â 9”‰d ’ù9 ‘©*‘ù@ ’y ”Ù “â &”y(”r

28. According to the notes, what is the ONLY type of wave that technically cannot exist in physics?

’‰K‘ùM“, ”r ’‰A“‰>“ Â -”É$“ù ” .”y •ù ’é>’IM“ *”Ù0‘Y>“   ” $“ ‘r ”É(“‰@ “™H ‘ÉK ’I ’‰@‘Y@ “ B’¢

Meaning:
Longitudinal Non-Mechanical (or Longitudinal Electromagnetic) wave.

•Y(” &”‰0”Ù'”Ù/ ‘yH“ Ù/“é ’IM“ ?‘R ‰/“â ’‰A’iH“ M’yM’ò 5“ù&”Ù/” $ ‘©A’éM’É ” /) ’I0•) –@

Trick / KD Hack
Logic Link: Electromagnetic fields intrinsically oscillate perpendicular (90°) to the direction of wave travel. Therefore, an EM wave is strictly Transverse. A longitudinal EM wave breaks the laws of electromagnetism.
Logic Link: “Y?’iM’ùA’B ” .”Ù,‘Y@’ò ”Ù7”y$”Ù0 •i ’I0“ù “ B’¢ 8”r $“ ‘r /“é$”Ù0“â ” &“ù6“â ”r 2•),“Y$ (90°) ’iK“)(‘Y

29. What general category of waves do Earthquake waves fall into?

’ÙB‘Y ’¢ $“ ‘yG•" $“ ‘yK•" ”   “ù8 “‰>’é>’‰M’ò 6”Ù0”y#” .”y •i$” 9”‰ ?

Meaning:
Mechanical Waves

’ù>•)$”Ù0“ù ’I0•) ”y (Mechanical Waves)

Trick / KD Hack
Logic Link: Earthquakes rely entirely on the Earth's crust and mantle (solid medium) to propagate. Because they require a medium to bump and shift, they are purely Mechanical.
Logic Link: ’ÙB‘Y ’¢ *”)0” $“ 9 “‰G ’¹H“)(”r ”r 2“ù ’©C’YM“Y@ ‘Y@ ’©*’ <”   “ .”y ‘ù2 (’ K“, .“é'”Ù/’â’ *“ (“ù0”Ù-“

30. What specific type of wave is a Surface Earthquake wave?

“‰$“™@ ’ÙB‘Y ’¢ …7W&`ace Earthquake) ’I0•) ‘Y?“, 5“ù6“ù7”Ù ’©M“ “é0 ‘Y@ ’I0•) “™H?

Meaning:
Longitudinal Wave (0 degrees).

•Y(” &”‰0”Ù'”Ù/ ’I0•) (0 ’ ?‘yM“ @)–@

Trick / KD Hack
Logic Link: Surface waves compress and expand the earth horizontally in the same direction the wave travels, acting just like a spring (0°).
Logic Link: “‰$“™@ ’I0•) ”y •™8” &“ù6“â .”y ’©C’YM“Y@ ‘YK ‘YM“yH’I?‘Â 0”)* “‰G “‰ ‘YA‘©?’B “ 5“ù8”Ù$“é

31. What specific type of wave is an Inner Earthquake wave?

•i ’I0“ù ’ÙB‘Y ’¢ „–ææW" V thquake) ’I0•) ‘Y?“, 5“ù6“ù7”Ù ’©M“ “é0 ‘Y@ ’I0•) “™H?

Meaning:
Transversal Wave (90 degrees).

•Y(” *”Ù0“‰M’R $“ ‘r …@ransversal Wave - 90 ’ ?‘yM“ @)–@

Trick / KD Hack
Logic Link: Inner waves displace the deep rock vertically (up and down) as the energy travels outward, forcing a 90° relationship.
Logic Link: •i ’I0“ù ’I0•) ”y ‘y9“ @ ‘© ”Ù “é(‘YK “) ’É5’B ‰ ’©0 ‘I0 ’‰@‘©G) “Y?“‰M’Y>’©?’B “ $” 9”‰ ‘YM’ùK•

32. The motion of a snake slithering or a rope being flicked represents what type of wave?

’¹?“‰2’IG “™A•ò 8“é ’¢ ” ’I? ’ù> ‘Ù ‘Y@ ‘É> “ 9” 0“‰M“‰@ ‘Y?“, *”Ù0‘Y>“ ” $“ ‘r ”² &“ M“i>’I@ “™H

Meaning:
Transverse Wave

•Y(” *”Ù0“‰M’R $“ ‘r …@ransverse Wave)

Trick / KD Hack
Logic Link: To go forward, a snake pushes its body side-to-side (90°). To send a wave down a rope, your hand moves up and down (90°). Both are perpendicular displacements.
Logic Link: •i ”r “é(”r ”r 2“ù , “‰>•)* •Y*’‰G “i0” 0 ‘YK •Y “"Ù,‘y2 (90°) ’y ”y2’I> “™H–B 0“‰M“‰@ “‰G “)9“ -”y